Quantum Numbers and Orbitals

Answer key with explanations

A. The quantum numbers

1. Meanings.

> **n**: the shell (size and main energy). **l**: the subshell (shape). **m_l**: the orientation of the orbital in space. **m_s**: the spin of the electron (+½ or −½).

2. n = 4.

> (a) l = 0 (**4s**), 1 (**4p**), 2 (**4d**), 3 (**4f**). (b) Orbitals: 1 + 3 + 5 + 7 = **16** = 4^2. Electrons: 2 × 16 = **32** = 2×4^2.

B. Allowed or not?

3. Allowed sets.

> (a) **Allowed**: a 1s electron.
>
> (b) **Not allowed**: for l = 1, m_l can only be −1, 0 or +1.
>
> (c) **Allowed**: a 3d electron.
>
> (d) **Not allowed**: m_s must be +½ or −½.
>
> (e) **Not allowed**: l must be at most n − 1 = 2.

4. Subshells.

> (a) **2p**, 6 electrons (b) **3d**, 10 (c) **5s**, 2 (d) **4f**, 14

C. Electrons in atoms

5. Possible sets.

> (a) $1s^1$: **(1, 0, 0, +½)** (or −½)
>
> (b) Potassium is [Ar] $4s^1$: **(4, 0, 0, +½)** (or −½)
>
> (c) 2p: n = 2, l = 1, m_l = −1, 0 or +1, m_s = ±½, e.g. **(2, 1, 0, +½)**. (By Hund's rule, carbon's two 2p electrons are in different orbitals with the same spin.)

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6. Pauli.

> All electrons in one orbital share the same n, l and m_l, so they can only differ in m_s. m_s has only two values (+½ and −½), so a third electron would have to repeat a full set of quantum numbers, which the Pauli exclusion principle forbids.

D. Shapes and nodes

7. Shapes and nodes.

> (a) s: a sphere centred on the nucleus. p_z: two lobes along the z axis, on either side of the nucleus, with a nodal plane (the xy plane) between them.
>
> (b) 2s: **0 angular, 1 radial**. 2p: **1 angular, 0 radial**. 3d: **2 angular, 0 radial**.

E. Challenge

8. Block widths.

> Each orbital holds 2 electrons, and each element in a block adds one more electron to that subshell. s has 1 orbital (2 electrons, **2 columns**), p has 3 (6 electrons, **6 columns**), d has 5 (10 electrons, **10 columns**) and f has 7 (14 electrons, **14 columns**). The numbers of orbitals come from m_l having $2l + 1$ values.

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